

Dhruv Kumar

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.39

Bhopal, India

2023 - 2027

EXPERIENCE

ShikshaLokam

Feb 2026 – Aug 2026

AI System Intern

- Contributed to the continued development and enhancement of the **Chanakaya Web Application**, designed to serve as a go-to academic support platform for teachers.
- Architected a **multi-agent backend** using **LangGraph** to dynamically route real-time teacher queries to **11 specialized AI tools**, powered by **Gemini 2.5 Flash** via a custom OpenRouter gateway.
- Engineered a retrieval pipeline utilizing **RAGFlow**, local **PostgreSQL embeddings**, and **LlamaParse** for complex NCERT document ingestion, integrating **Sarvam AI** to process and translate classroom speech across **12+ regional Indian languages**.
- Executed the migration from **Supabase** to a **self-hosted PostgreSQL** database using a custom async ORM, and developed **deterministic scoring engines** to optimize personalized student questioning and engagement tracking.

PROJECTS

Guru Sikshan | *MFE, FastAPI, RAG, PostgreSQL, Gemini*

Feb 2026

- Built an **AI-powered platform** where DIET admins manage teacher rosters and upload training modules for personalized professional development, later scaled to production by **ShikshaLokam**.
- Engineered the **backend API layer** linking a React Native frontend to a **RAG pipeline** utilizing **RAGFlow** and optimizing prompt design to improve response accuracy by **80%**.
- Designed a **feedback intelligence system** that summarizes **100+ raw responses** per module and assigns attention scores, converting noisy feedback into **prioritized, actionable issues**.
- Implemented a secure **API gateway** with session management and containerized the stack using **Docker** for reproducible deployment.

RoadIQ | *Python, YOLO11m, BoT-SORT, FastAPI, OpenCV, Shapely*

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- Developed a real-time **Computer Vision pipeline** using **YOLO11m** and **BoT-SORT** to accurately track vehicles and passengers, implementing **custom filtering** to reduce false-positive detections.
- Engineered **real-time detection engines** for **four traffic violations** (illegal parking, wrong-side driving, missing helmets, and triple-riding) utilizing **spatial data** and **trajectory analysis**.
- Built an automated **citation generator** and a **FastAPI-backed analytics dashboard** for real-time monitoring and **data export**.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, SQL, JavaScript

Frameworks & Libraries: numpy, pandas, FastAPI, Flask, LangGraph, OpenCV, Scikit-learn

AI/ML: Agentic AI, RAG, NLP, LlamaParse, Sarvam AI, Sentence-Transformers, Deep Learning, Computer Vision

Infrastructure & Tools: PostgreSQL, Supabase, AWS, Docker, Git

ACHIEVEMENTS & CERTIFICATIONS

- Hackathon Winner:** Secured **1st place** and a **Rs. 1.5 Lakh prize** among **2000+ teams** at the I4EE Hackathon (InvokED 5.0) by HackerEarth and ShikshaLokam by developing Guru Sikshan, an AI-powered platform for personalized teacher training.
- Open Source Contributor:** Selected for GirlScript Summer of Code (GSSoC) 2026.
- Certifications:** Market Analytics (NPTEL), Google IT Support Certificate, The Bits and Bytes of Computer Networking (Google), Introduction to Machine Learning (NPTEL).